

E-Commerce Performance & Logistics Optimization

Bootcamp Data Analyst with SQL &
Python using Google Platform

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Business Questions & Goals



Market Concentration

Which provinces and cities are the biggest revenue contributors?



Purchasing Power

How is the Average Order Value (AOV) distributed across Indonesia?



Logistics Popularity

What are the most used shipping options, and their fee impact?



Cost Efficiency

Which regions show the highest shipping gap (subsidies) inefficiency?



Optimize Growth Strategy

Identify high-revenue provinces to maximize marketing budget allocation and geographical expansion.



Refine Logistics Efficiency

Detect significant shipping variances to implement cost-reduction and localized hubs strategy.



Enhance Operational Insights

Analyze regional purchasing power and transaction behavior to enhance product strategy.

Source

Public E-Commerce Dataset sourced from Kaggle, simulating Indonesian market transactions.

Coverage

18,000+ raw observations filtered into 16,000+ successful transactions across 38 provinces.

Variables

15+ features including order intelligence, sales performance, geographic demographics, logistics efficiency, and customer behavior.

Tools & Libraries

1. Python (pandas, numpy, matplotlib, and seaborn)
→ Data cleaning & visualization
2. Looker
→ Interactive BI dashboard

Preparation

Data Cleaning



Missing Value Imputation

Handled nulls using median for numerical data and constant labels for categories.



Text & Date Normalization

Standardized mixed date formats and unified payment/logistics alias inconsistencies.



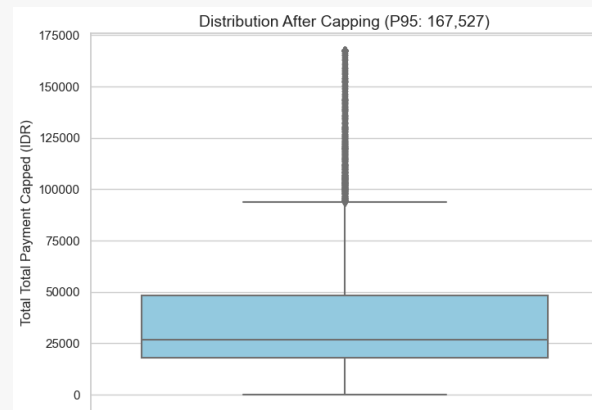
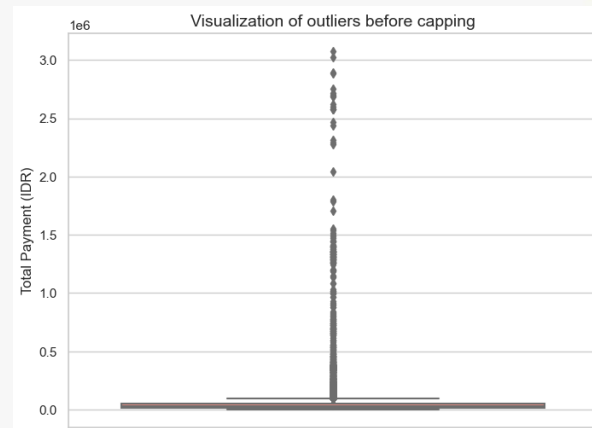
Outlier Capping (P95)

Applied Percentile 95 capping on Revenue to stabilize distribution for accurate metrics.



Feature Engineering

Created Shipping Gap and simplified Product Categories for deeper strategic insights.



Exploratory Dashboard

Geographic Revenue Distribution

“Sales are highly concentrated in the metropolitan areas of Java, with Jabodetabek emerging as the primary growth engines.”

✓ **Regional Domination:** West Java, Banten, & Jakarta account for over 55% of total revenue share.

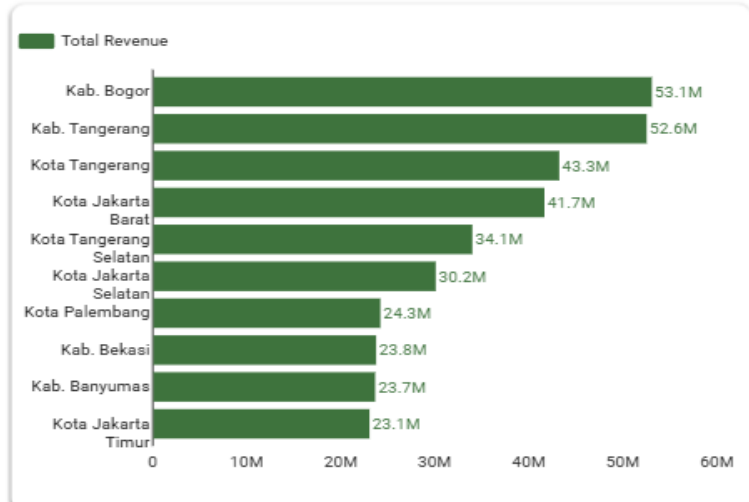
✓ **Market Leaders:** Bogor and Tangerang are the top GMV contributors.

✓ **Expansion Potential:** Emerging hubs in Palembang show potential for localized marketing campaigns.

Revenue Distribution by Geography



Top 10 Cities by Total Revenue



Exploratory Dashboard

Purchasing Power Analysis

"While Java dominates in transaction volume, Outer Island regions lead the national Average Order Value (AOV), with Papua, Sulawesi, and Kalimantan emerging as the highest-spending markets per transaction."



Outer Island Supremacy: The top 10 highest AOV rankings are dominated by regions outside Java, led by Papua and Sulawesi.

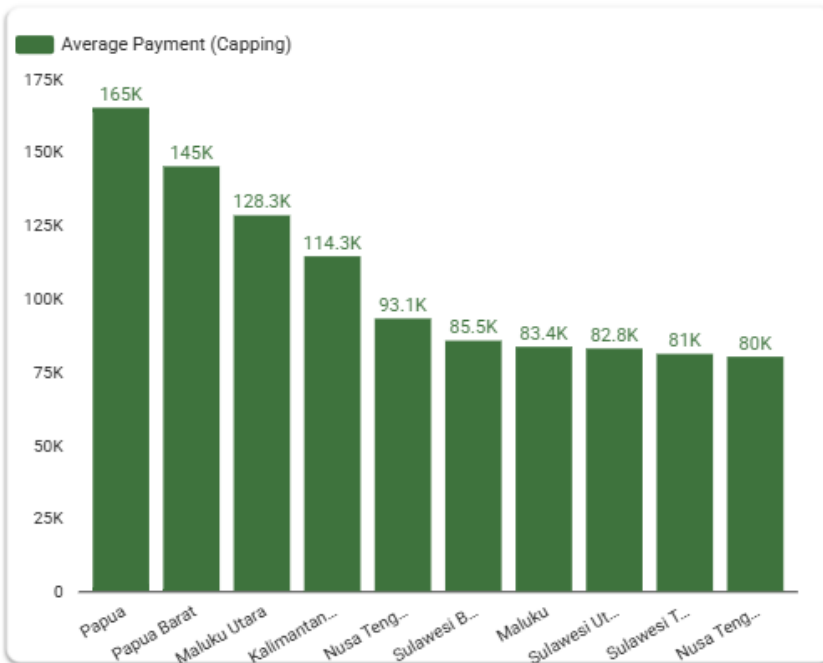


Spending Behavior: Customers in remote areas tend to have larger basket sizes, contrasting with the high-frequency/low-ticket model in Java.



Growth Opportunity: High AOV regions represent significant revenue potential for premium categories and wholesale-style marketing.

Average Order Value (AOV) per Province



Note: Top 10 provinces are exclusively located outside Java Island.

Exploratory Dashboard

Logistics Popularity

"Logistics choices are heavily driven by cost-efficiency, with the 'Hemat' service dominating 63% of the market share as customers prioritize low shipping overheads over delivery speed."



Budget Dominance: Over **86%** of the market share is captured by Hemat and Reguler Spx options, highlighting extreme price sensitivity.

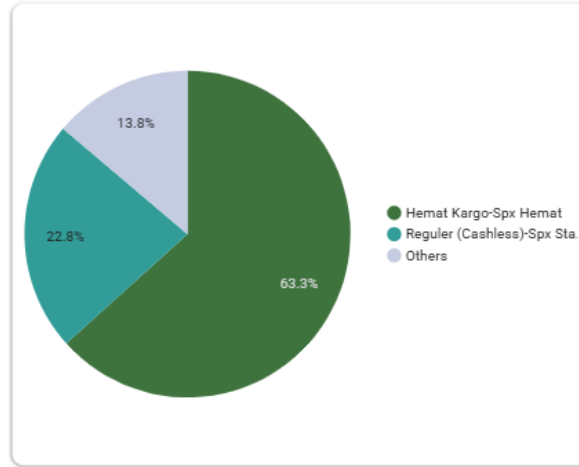


Cost Disparity: Average fees vary drastically from IDR 922 (Reguler) to IDR 25k (Kargo), dictating user preference for the 5k 'Hemat Kargo' sweet spot.

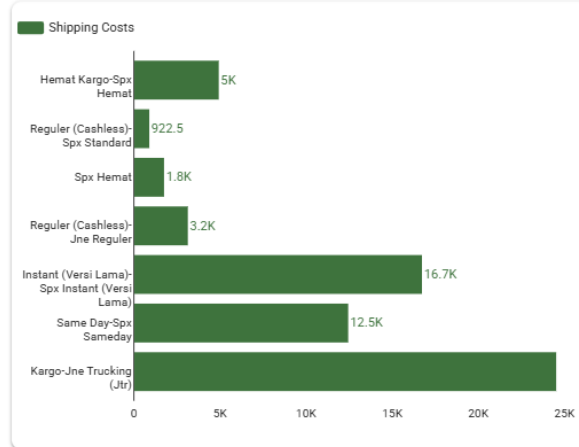


Premium Segment: Instant and Same-Day services serve a high-value niche (IDR 12k-16k), representing urgent urban delivery needs.

Shipping Carrier Distribution



Average Shipping Cost by Delivery Option



Exploratory Dashboard

Logistics Inefficiency

"Eastern Indonesia faces the highest logistics friction, requiring subsidies of up to IDR 47.3K per transaction to maintain market accessibility."



High Subsidy Burden

Remote provinces like Maluku and Papua show a cost gap nearly 5x higher than urban hubs, signaling major operational friction.



Margin Compression

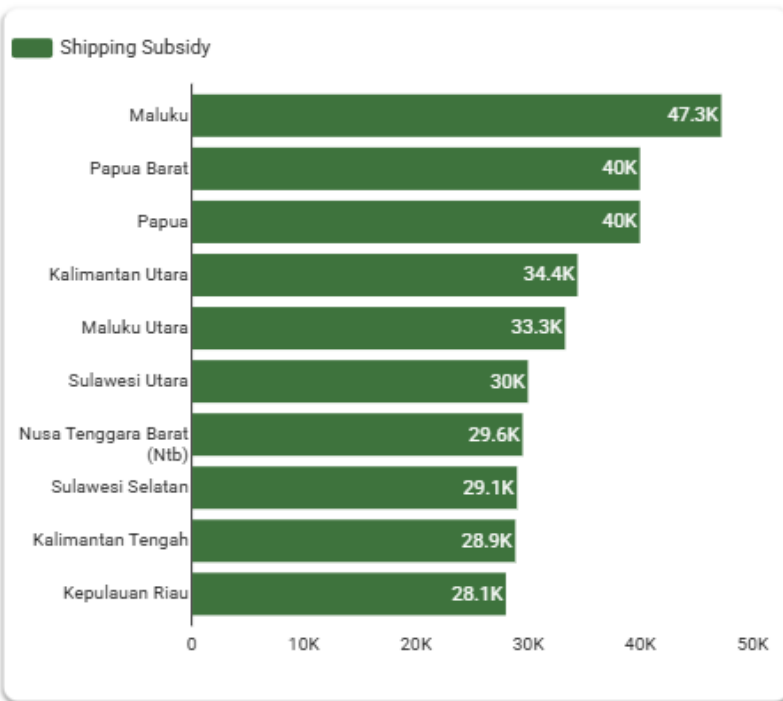
Average subsidies of ~40K per order in Eastern regions significantly offset the high AOV gains identified in previous analyses.



Optimization Strategy

Implementing a localized warehousing (Hub-and-Spoke) model in remote hubs is critical to mitigate long-haul logistics costs.

Average Shipping Cost Gap by Province



Product Strategy : Pareto Analysis

"The 80/20 principle holds true: a small subset of products drive the vast majority of e-commerce revenue."



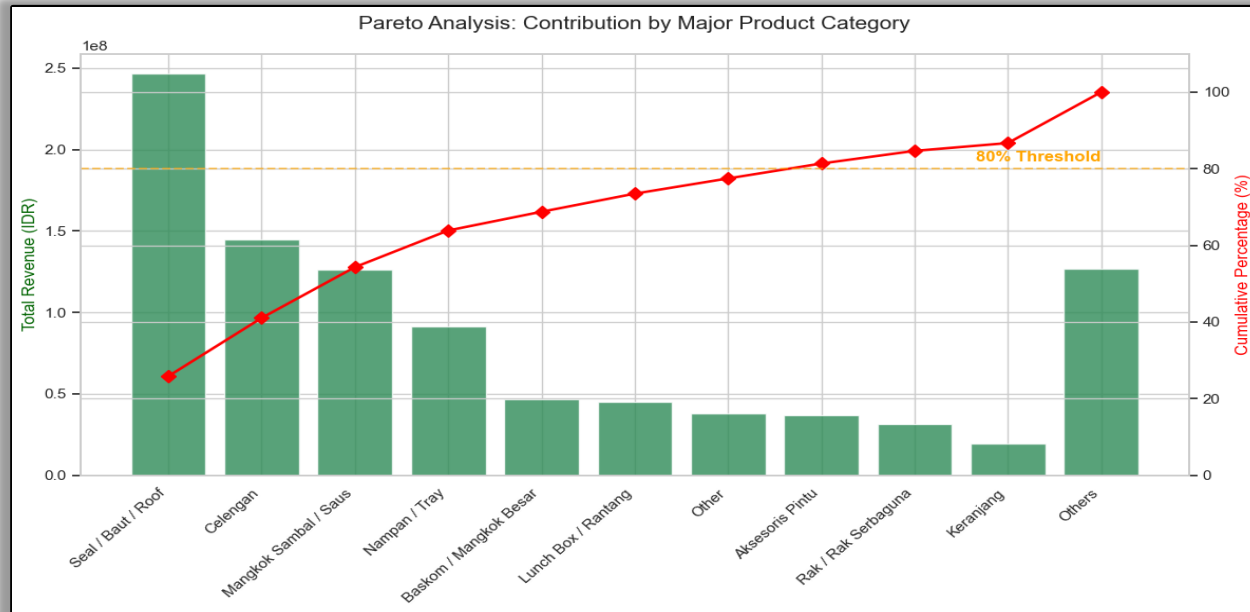
Revenue Concentration: 20% of categories generate roughly 80% of revenue, identifying the "Vital Few" for business growth.



Stock Optimization: Prioritizing these top-tier categories like seal/baut/roof, celengan, and mangkok sambal.



Inventory & Cost Rationalization: Apply Lean Inventory for the 'Useful Many' (bottom 80%) categories and consolidate shipment.



"Nearly 43% of cancellations are driven by buyers needing to modify their initial purchase decisions."



23.38% - Changed Mind: The leading driver of churn, highlighting high price-sensitivity and impulsive behavior.



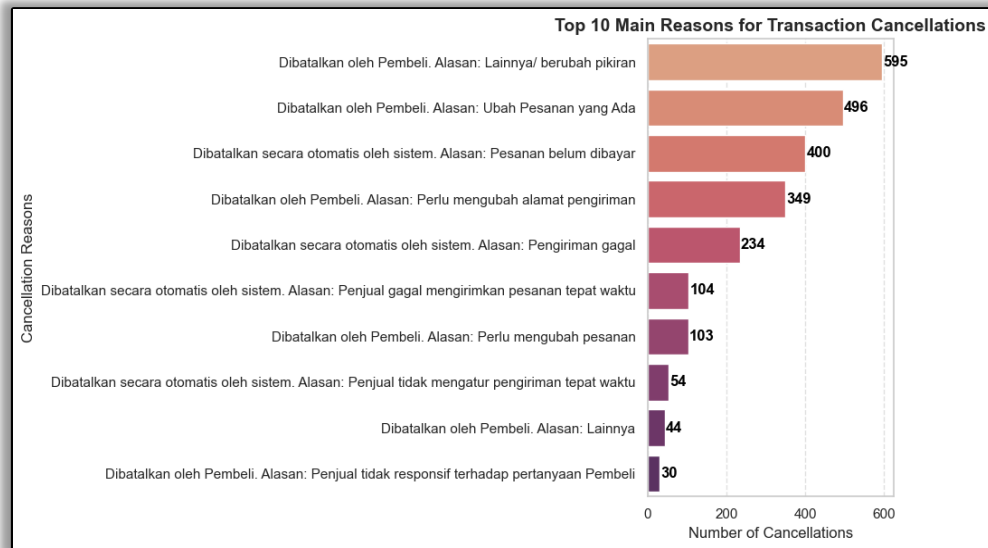
23.54% - Order Updates: Combined reasons for modifying orders indicate a major need for post-checkout item editing tools.



15.72% - Unpaid Orders: A significant portion of lost revenue stems from payment friction or abandoned checkouts.



13.71% - Address Change: Logistical input errors remain a core operational hurdle that requires UI/UX validation.



"Promotions drive volume, but subsidies impact revenue. Strategic discounting shows the strongest link to sales growth."



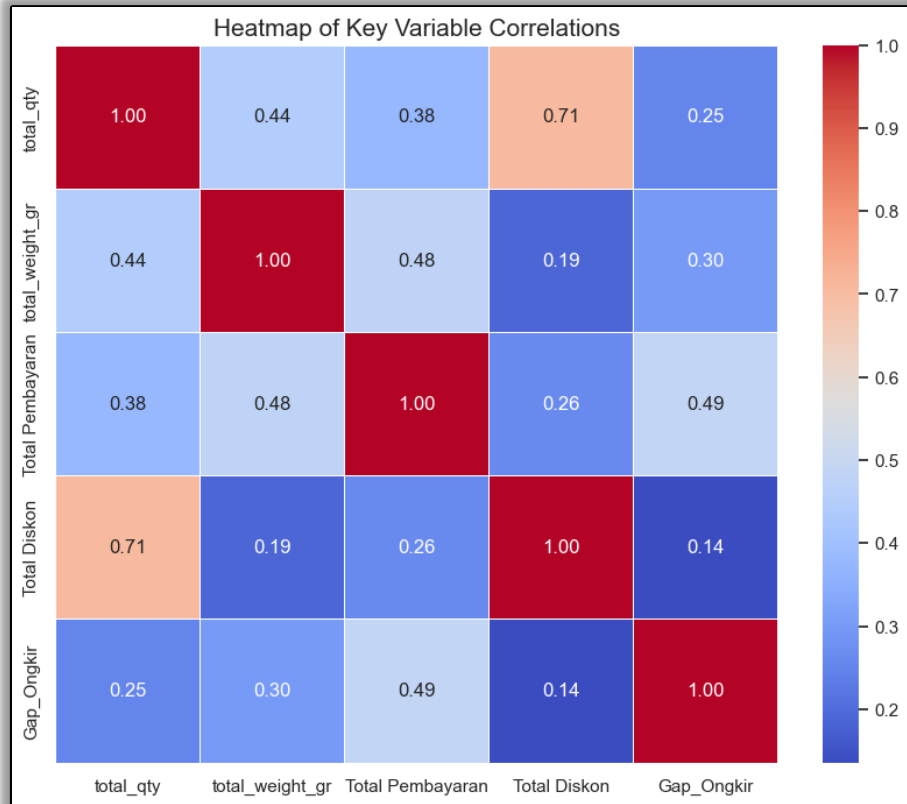
Promo Power (0.71): Strong correlation between Discounts & Quantity. Proven: Promotions are highly effective at driving bulk purchases.



Revenue Subsidy (0.49): Moderate correlation between Revenue & Shipping Gap. Insight: High-value orders often rely on heavy shipping subsidies.



Weight Factor (0.30): Low-moderate link between Weight & Shipping Gap. Indicates that subsidies are likely driven by regional zones, not just item bulkiness.



CONCLUSION

Growth Dynamics

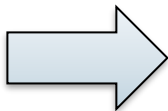
Volume is highly elastic to discounts (0.71 corr). However, revenue gains are currently offset by high logistics overheads in remote regions.

Logistics Friction

Eastern Indonesia remains the highest operational burden with shipping gaps reaching IDR 47K/order, eroding profitability despite high AOV.

Cancellations Churn

~43% of GMV leakage is driven by post-purchase revisions (Change of Mind/Address). This indicates a friction-heavy checkout-to-fulfillment flow.



RECOMMENDATION



Marketing Allocation

Allocate 60% of the ad budget to Java for volume growth and 40% to Eastern regions to target high-value product segments.



Logistics Partnership

Secure specialized volume-based contracts with economy-tier carriers to reduce shipping overhead without compromising customer reach.



Customer Retention & Churn Mitigation

Focus on Repeat Orders and implement a 15-minute Edit Order window to reduce the 43% cancellation rate caused by address/item revisions.



Fulfillment Optimization

Transition from direct subsidies to investing in Micro-Fulfillment Hubs in Eastern regions, enabling localized delivery and margin recovery.

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DASHBOARD LINK:

<https://datastudio.google.com/reporting/64f8de1e-82ac-49bf-b764-2e0cf58b108d>

LINK PYTHON:

https://github.com/Maulanafajaraji/indonesia-ecommerce-analysis/blob/main/indonesia_ecommerce_analysis.ipynb

Thanks!